

Advanced Computer Graphics

Motion Retargeting II

Example 1:

M. Gleicher,
Retargetting Motion to New Characters,
published in SIGGRAPH 1998

Introduction

- Retargetting motion
 - Adapting an animated motion from one character to another
- Mainly focus on motion retargetting to another figure with the same structure
- Pragmatic approach
 - Aim to preserve as many of the desirable properties of the original motion as possible

Motivation

- Motion data reusability
- Pragmatic approach
 - Limited ability to define high-level qualities of the motion mathematically
 - Limited ability to compute adaptations of complex metrics efficiently

The retargetting method

- The desirable properties are enforced as constraints
- Adaptation
 - If constraints are violated when motion is applied to a different figure, re-establish them to fit with the motion
- Spacetime constraints method
 - Considers the entire motion simultaneously
- Preserving the qualities of original motion
 - Minimize the changes & restrict frequency content

Previous work

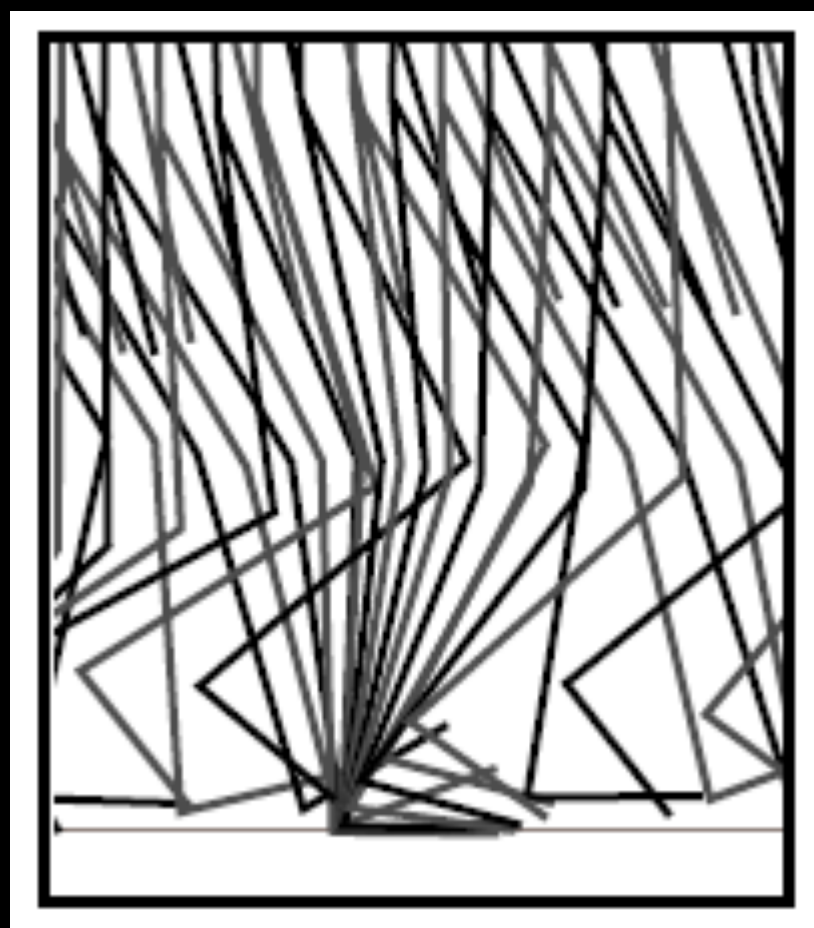
- Retargetting by manually tweaking each frame
 - Kinetix's Character Studio (1997)
- Motion re-generation
 - Adjust parameters of physical simulation to adapt controller for use with a new character or a character that is changing
 - Risks losing qualities in the original
 - Hodgins and Pollard (1997)
- Background for recent interest in motion alternation tools
 - Treat animated motion as time-varying signals and apply signal processing techniques
 - Litwinowicz's Inkwell system (1991), Motion blending by Perlin (1995), Motion warping by Witkin and Popovic (1995), etc.

Spacetime Constraints Method

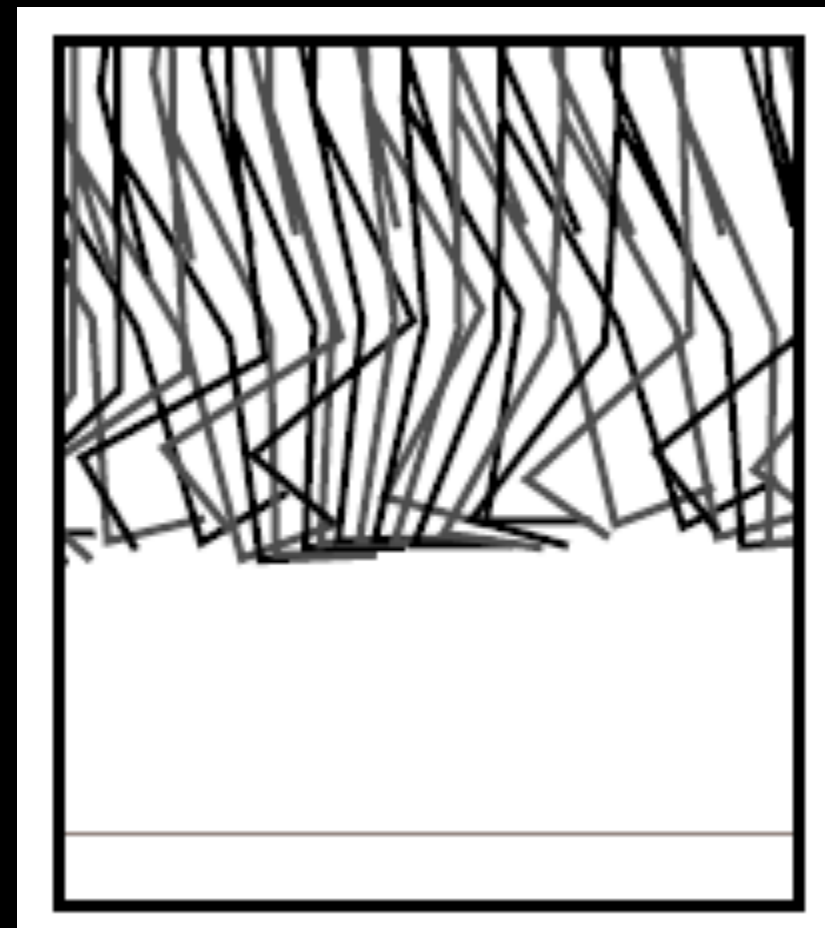
- Poses the motion synthesis problem as a constrained optimization
 - What is the best motion that meets a set of constraints?
- Unlike other constraint methods, it poses a single large problem over a duration of motion, rather than on an individual frame
- Gives “global view” to consider relationships among multiple frames

Failure case (1)

- $m(t) = m_o(t) + d(t)$
 - $m_o(t)$: original motion, $m(t)$: retargetted motion
 - $d(t)$: distance between two motions
- Target character fails to interact correctly with other objects by simply reusing the original motion



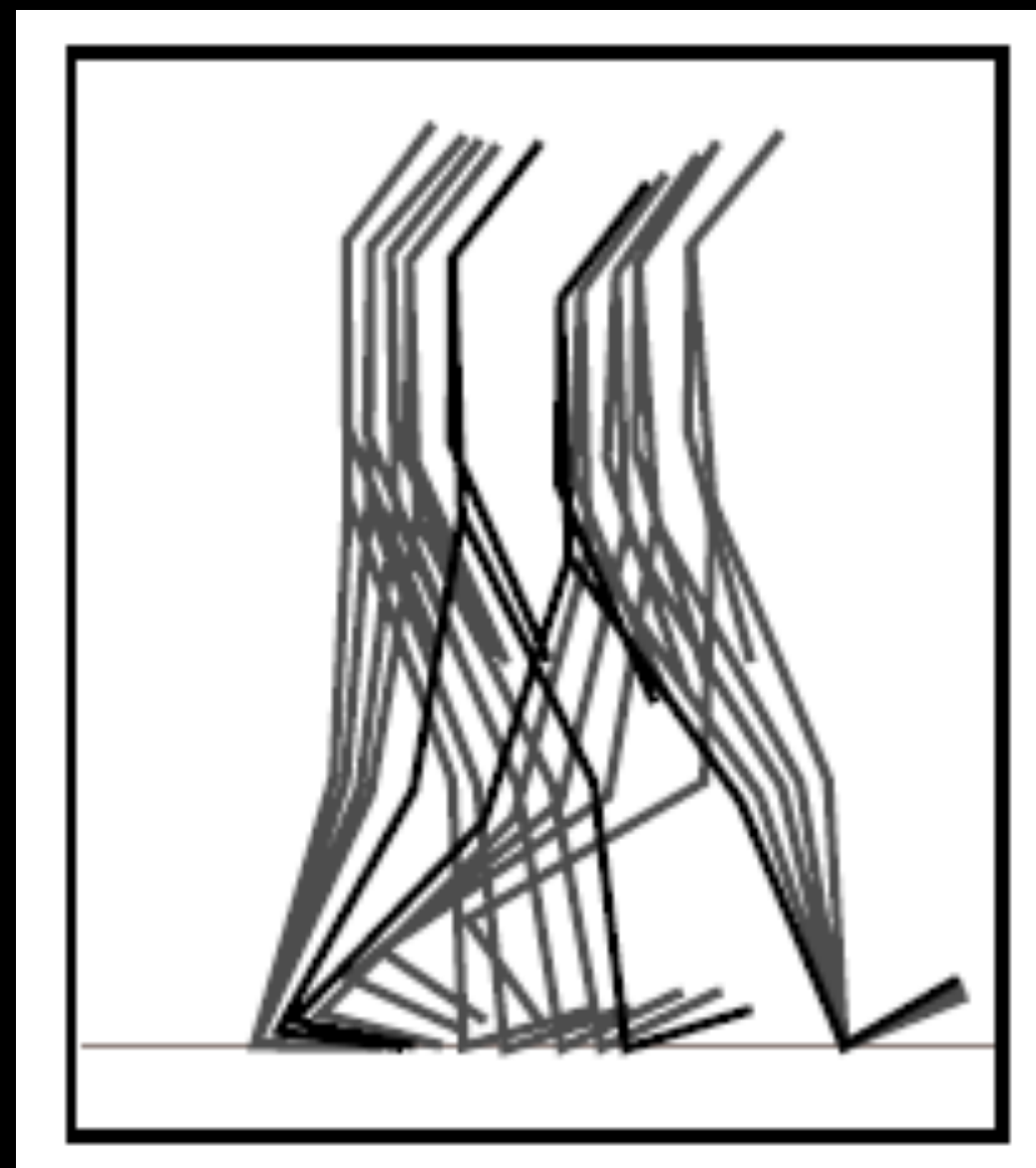
Original motion



Retargetted motion

Failure case (2) Inverse Kinematics

- IK solver adjusts the configuration of the character to meet the constraints in each frame (ex. feet on the floor)
- Inconsistency in frame-wise adjustment
 - Each frame is altered differently to meet the constraints
 - High frequency “jerkiness”



Noticeable discontinuity in IK-applied motion

Failure case (3) Low-pass filtering

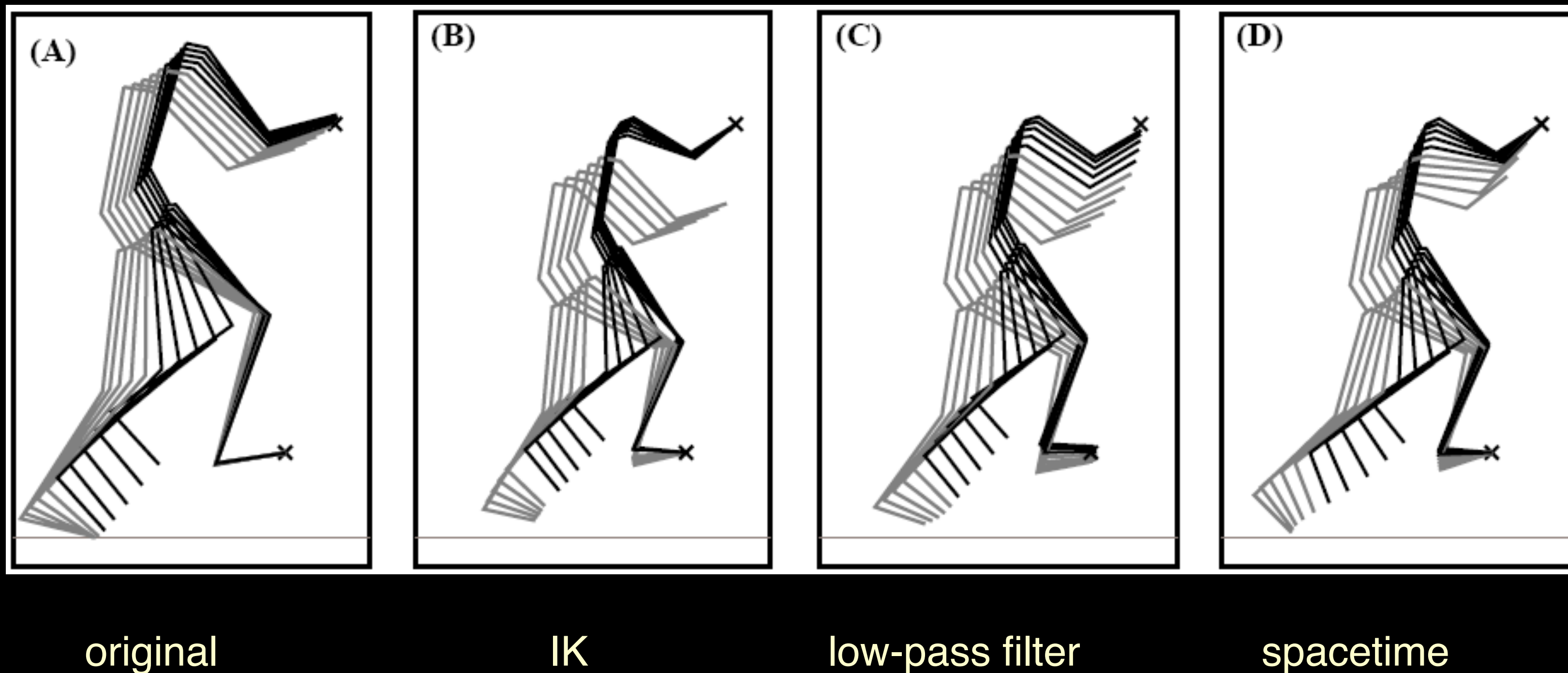
- High frequency added by IK caused problem
- Low-pass filter the signal to avoid such additional high frequencies



Snapes are removed at the expense of violating the constraints

Comparison of results

- Extreme high frequencies are undesirable, but it's still important to preserve high frequency contents
- Failures of per-frame approach motivate spacetime constraints



Spacetime revisited

- Look at the entire motion, and make choices based on other parts of motion
- Poses retargetting problem mathematically to use numerical methods to solve the constrained optimization problem
 - Seek a motion $m(t)$, satisfying $f(m(t)) \diamond c$
 - $f(\cdot)$: constraint function, c : constant, and
 - $\diamond \in \{\leq, \geq, =\}$
 - that minimizes an objective function $g(m)$
 - $g(m) = \int_t (m(t) - m_o(t))^2 = \int_t d(t)^2$

Spacetime in practice (1)

- Difficulties in realizing the spacetime ideal for retargetting
 - Some properties are difficult to encode mathematically
 - Not all the properties required are known
 - In given setting, we must decide which properties are important
 - Many properties and constraints are specific only to small set of examples
- Even if the desired animation is encoded completely, finding solution using such rich sets of constraints can be too difficult

Spacetime in practice (2)

- Use pragmatic tools to define a spacetime problem with desired solution
 - Constraints
 - Specific features of motion that must be maintained
 - Objective function
 - To limit certain generally unacceptable types of changes
 - Representation
 - To solve the optimization problem effectively
 - Starting point
 - For constrained optimization

Sources of constraints

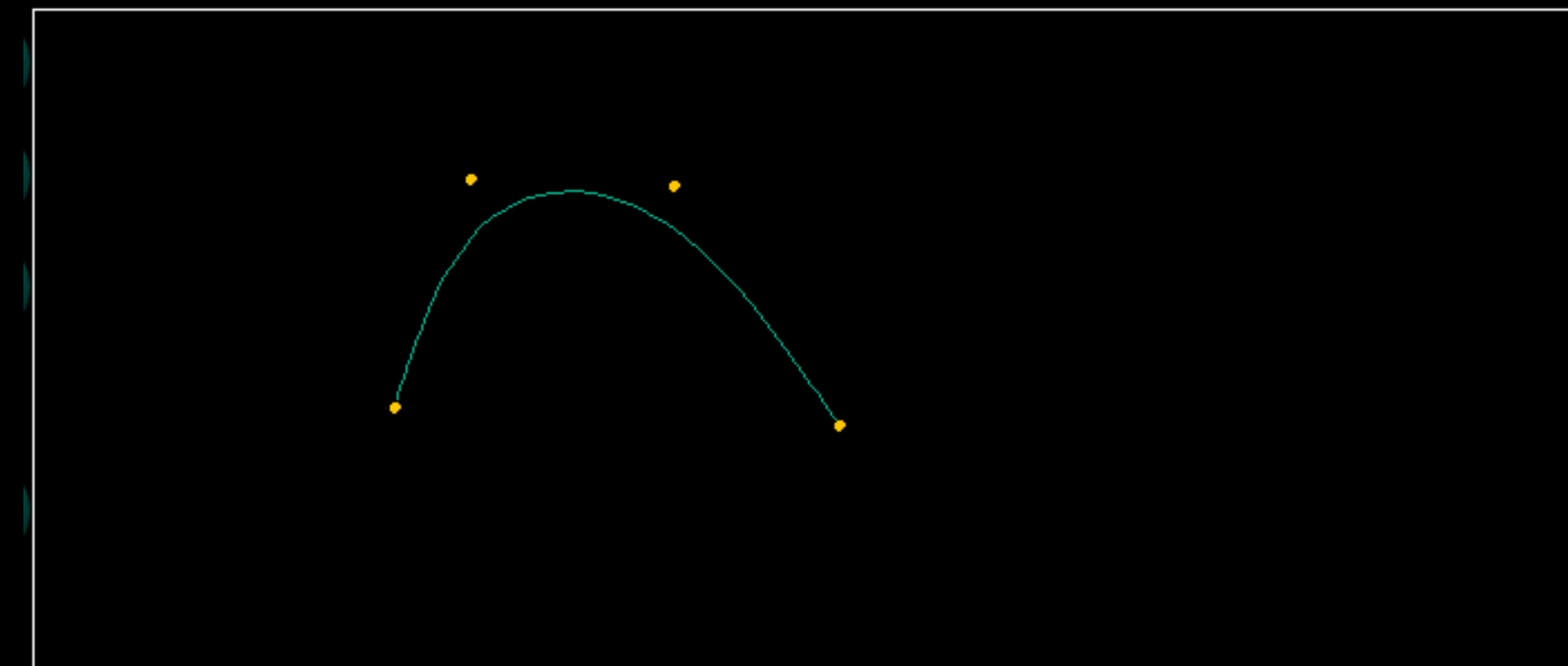
- Constraints identify features of original motion that must be preserved in retargetted result
 - Defined once for each motion, and used for any retargetting
- Mathematically, differentiable functions of the parameters of the character ($f(\cdot)$)
- When making a new constraint, it must be invariant of other aspects of the motion
- Examples
 - A parameter's value is in a range
 - A point on the character is in a specific location
 - A point on the character is in a certain region

Objective functions

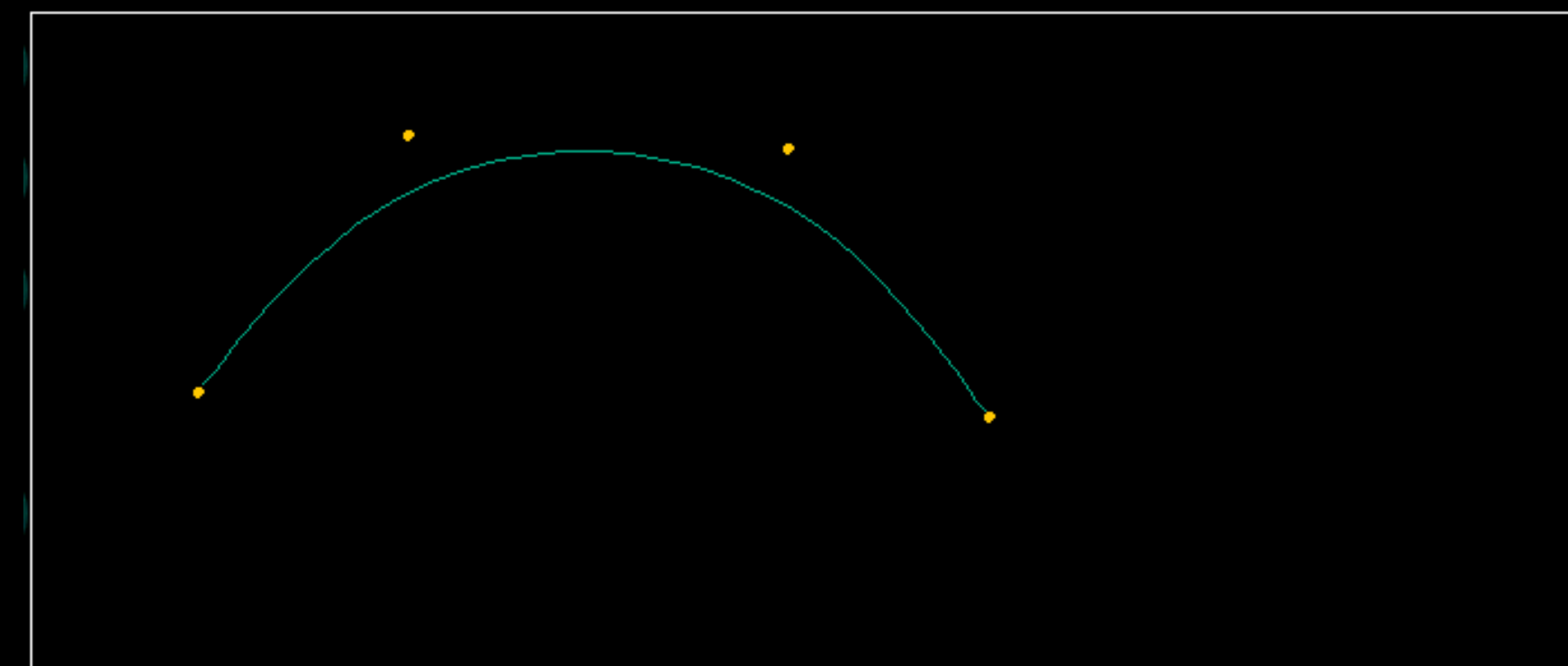
- Among many possible motions that satisfy constraints, we need to select the best choice
- Simple objective for retargeting is:
 - Minimize the amount of noticeable change
 - $g(m) = \int_t (m(t) - m_o(t))^2 = \int_t d(t)^2$
- Effectively preserves high frequency content

Representation (1)

- To solve optimization problem effectively, we need to restrict high frequency content of the changes
- Motion-displacement map
 - $m(t) = m_o(t) + d(t)$
- Use cubic B-splines with control point spacing determined by the desired frequency limits
 - Different key spacing for different parameters



Narrow control point spacing



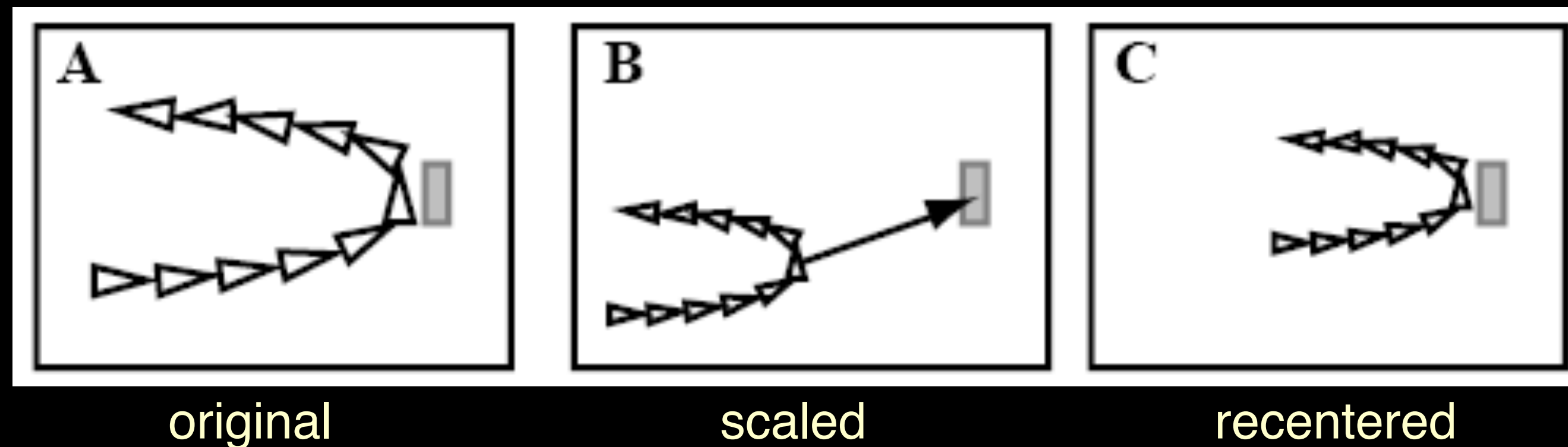
Wide control point spacing

Representation (2)

- Use bandpass decomposition and choose the key spacing that coincides with the highest frequency
 - Uniform key spacings of 2, 4, and 8 could be used, and the one is chosen that gives the best appealing result
- If no exact solution to the constraints is found, choose the one that minimizes the residual of the constraints

Starting points (1)

- Positional offsets are not scale-independent
 - Especially when the character interacts with world
- First, multiply the positional parameters to scale the motion around an arbitrary point
- Then, add a transitional component to them to recenter the scaling of positional parameters



Starting points (2)

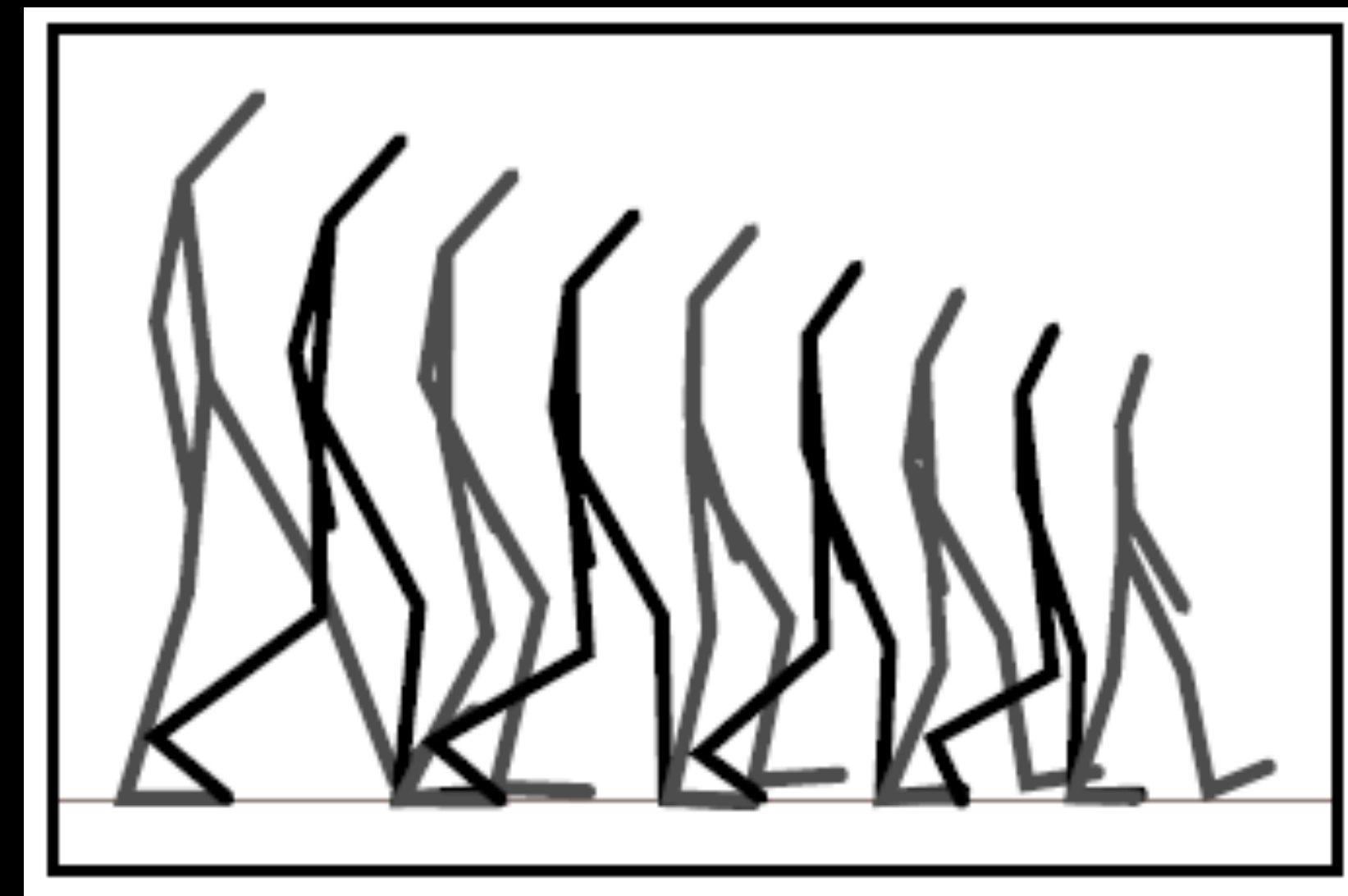
- For translation, simple shift of the motion will do
 - Constant positional shift of motion is not noticeable, except when the character interacts with world
- We can get the shift by computing average displacement
 - Displacement means the vector between the point on character and the position it's attached to
- Again, we can get undesirable high frequencies because center of scaling can vary over the whole motion
 - Interpolate the offsets to frames without any displacement, and low-pass filter to remove high frequencies

Algorithm overview

1. Begin with initial motion $m_1(t)$ with identified constraints
2. Find an initial estimate
3. Choose representation for motion-displacement curve based on frequency decomposition of original motion
4. Solve non-linear constraint problem for a displacement to provide motion satisfying constraints
5. If the result is not satisfactory, solve again with initial motion $(m_1(t)+d(t))$, with denser set of control points

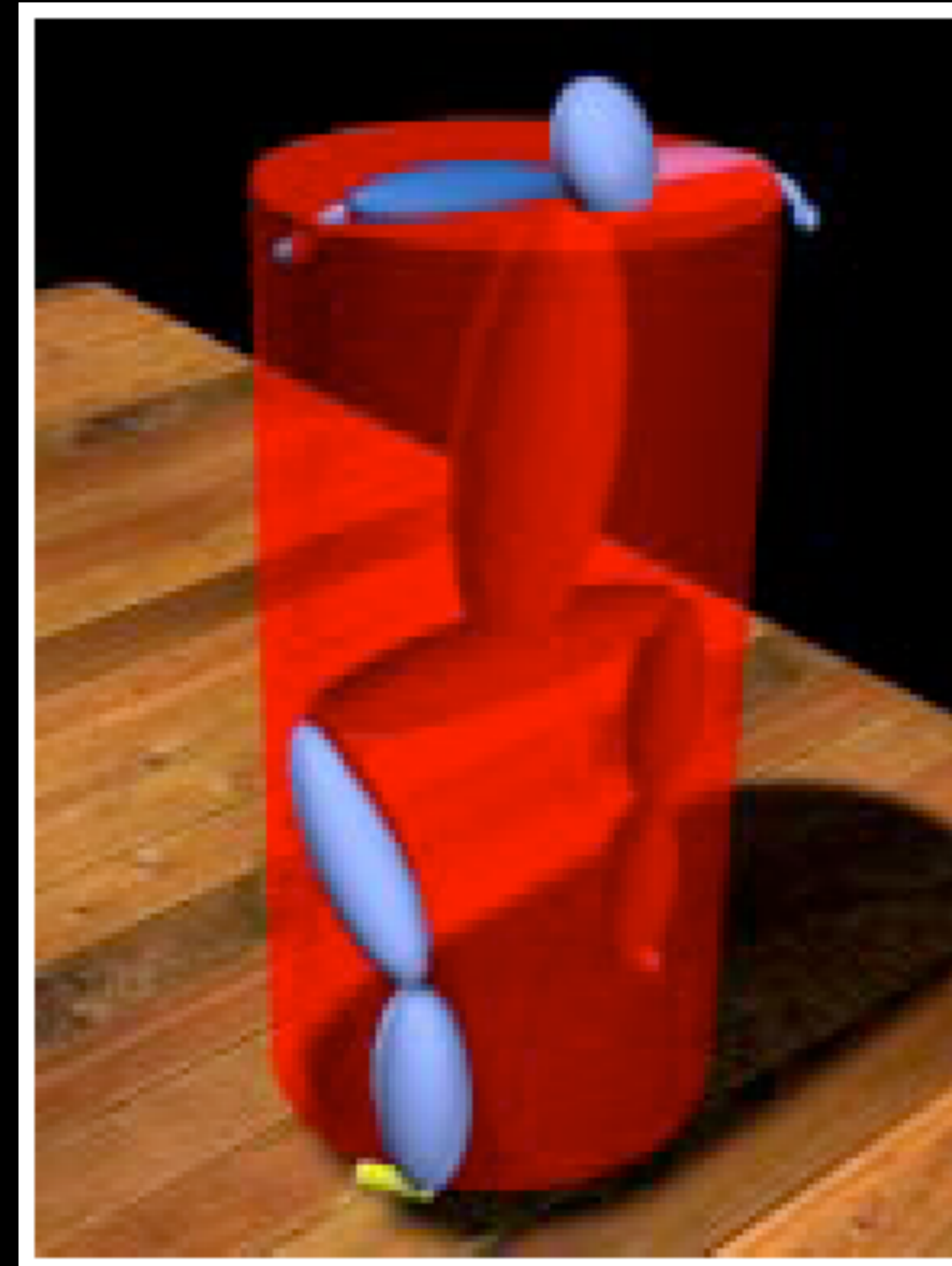
Motion morphing

- Use the same methods as in retargetting
- Only difference is that for morphing, the segment lengths of the target character is not constant
- We can use a different scaling amount on each frame in Step 2 of above algorithm



Differing characters

- Target character has different structure
- User identifies correspondences between externally visible features of the original & target characters
- Then, use the same spacetime constraints technique



Solving non-linear optimization (1)

- Constrained optimization problem is generically:
 - Minimize $g(x)$ subject to $f(x) = c$
 - We need to find the values for the B-spline control points for each parameter
 - x : concatenation of control points
 - Here, we only consider equality constraints
- We can approximate equation ' $g(m) = \int_t (m(t)-m_o(t))^2 = \int_t d(t)^2$ ' as:
 - $g(x) = \frac{1}{2}xMx$, M : diagonal matrix (denotes weights of each parameter)
- We can construct B-spline of each parameter individually because complicated individual constraints can be decomposed into smaller pieces

Solving non-linear optimization (2)

- As it is unreasonable to expect equational constraints, an alternative solution approach can be:
 - Minimize constraint residual r s.t.
 - $r = \frac{1}{2} \cdot (f(x) - c) \cdot (f(x) - c) + \epsilon \cdot \frac{1}{2} \cdot x \cdot x$
 - $\epsilon \cdot \frac{1}{2} \cdot x \cdot x$: constraints to fully determine the solution
- Non-linear least-squares solver iteratively improves on an estimate of the solution
 - For each step, linear approximation of the constraint problem
 - $f(x + \Delta) \approx f(x) + \frac{\partial f}{\partial x} \Delta$, : Taylor expansion
 - $\Rightarrow J \Delta = f(x) - c$, : J = Jacobian matrix
 - $\Rightarrow (J^T J + \epsilon I) \Delta = J^T (f(x) - c)$: damped pseudo-inverse
 - $\Rightarrow$ we can find Δ

Solving non-linear optimization (3)

- Once Δ is found, we determine a value of k s.t. $x+k \Delta$ best satisfies the non-linear constraints
- This least-squares solver turned out to be faster than SQP style solvers in most cases

Example 2:

K. Aberman, P. Li, Dani Lischinski, Olga Sorkine-Hornung, Daniel Cohen-Or,
Baoquan Chen,
Skeleton-Aware Networks for Deep Motion Retargeting,
presented in SIGGRAPH 2020

Skeleton-Aware Networks for Deep Motion Retargeting

Kfir Aberman*, Peizhuo Li*, Dani Lischinski, Olga Sorkine-Hornung,
Daniel Cohen-Or, Baoquan Chen

(The video contains voice)



ETH



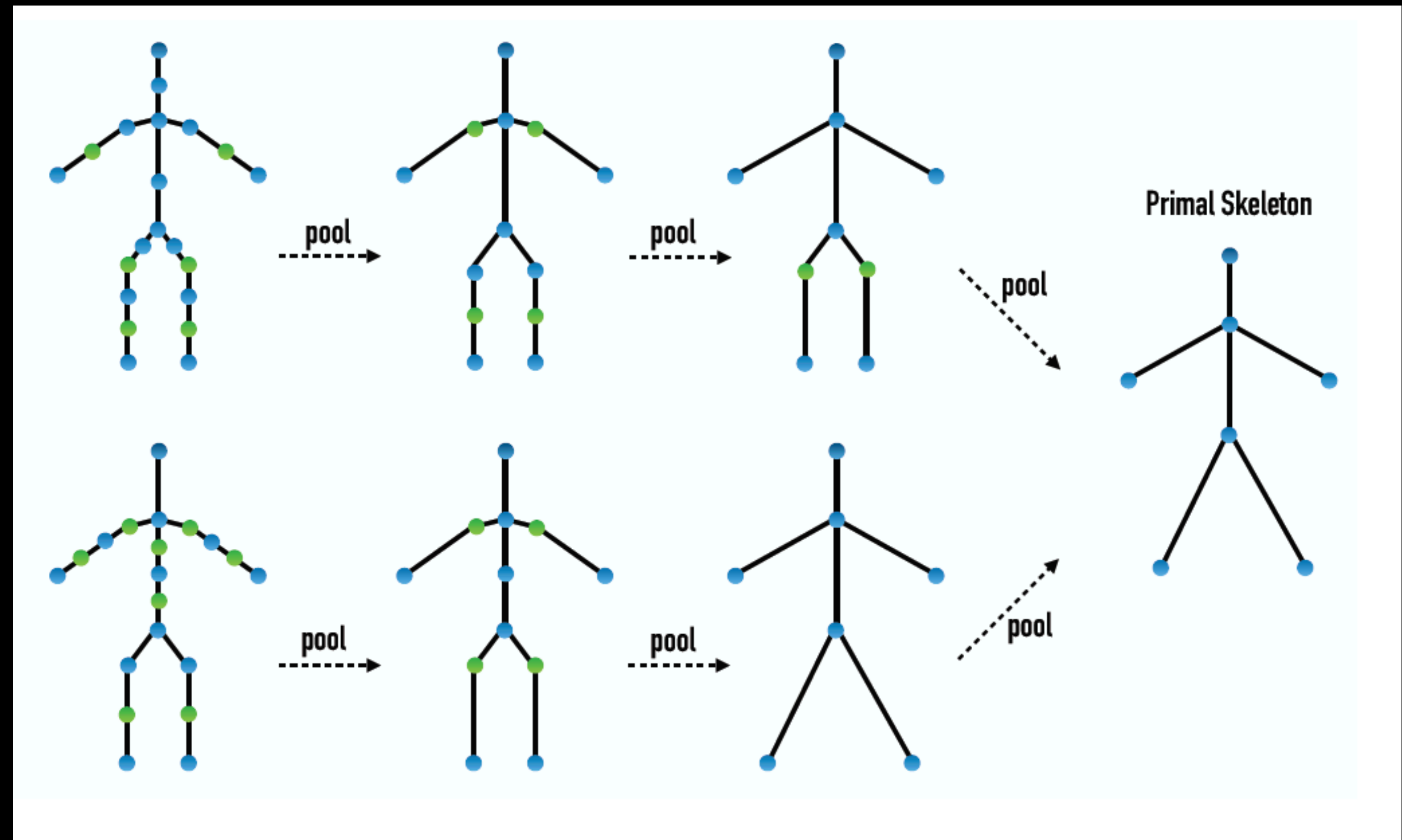
SIGGRAPH THINK
BEYOND
2020 19-23 JULY WASHINGTON DC

Background

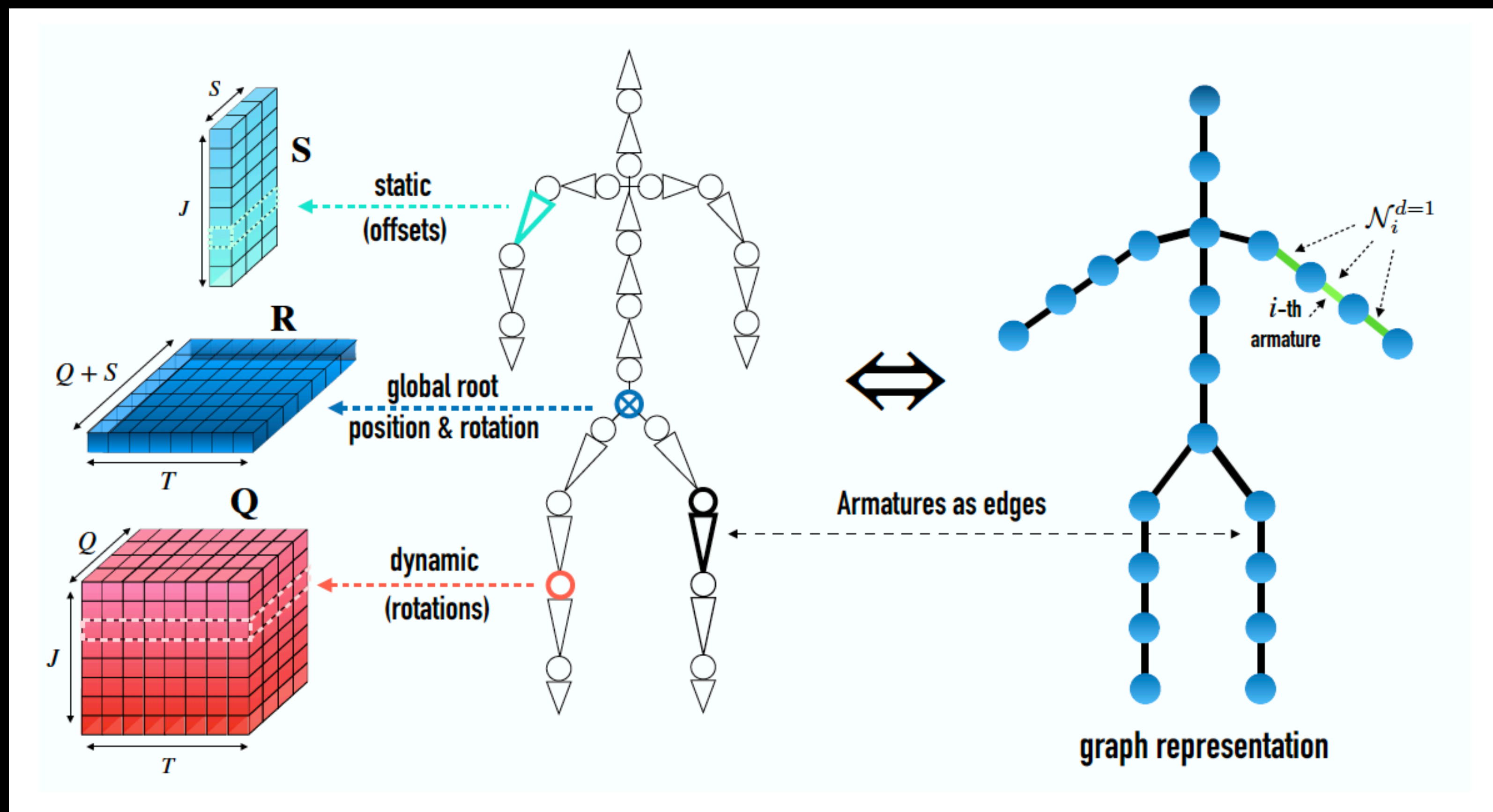
- Retargetting problem
 - Finding correspondences
- Approach
 - Find it using deep-learning methods

Approach

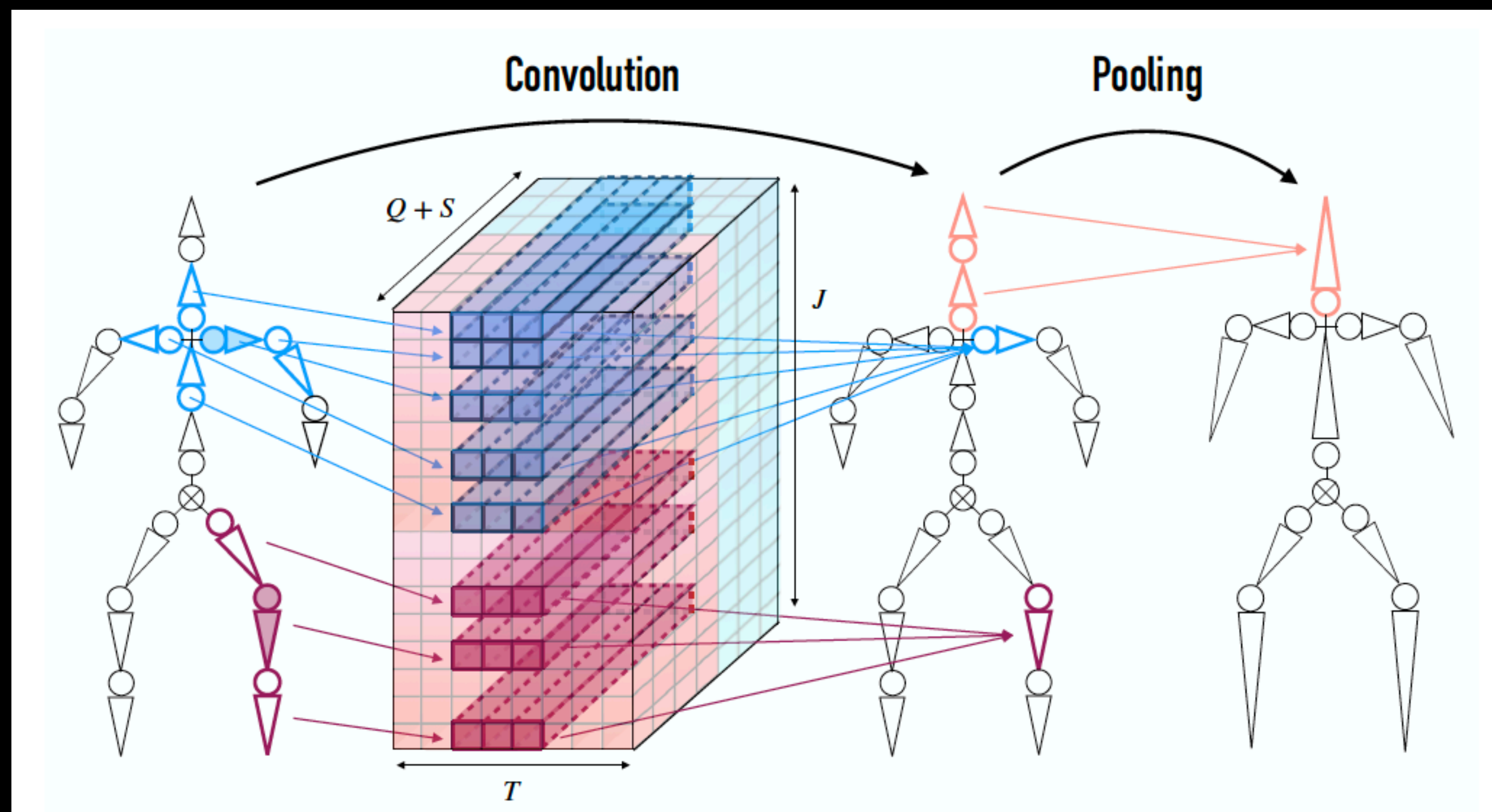
- Different structures
 - Make a simplified common structure
 - Using deep-learning
 - Create a map to reconstruct detailed motion



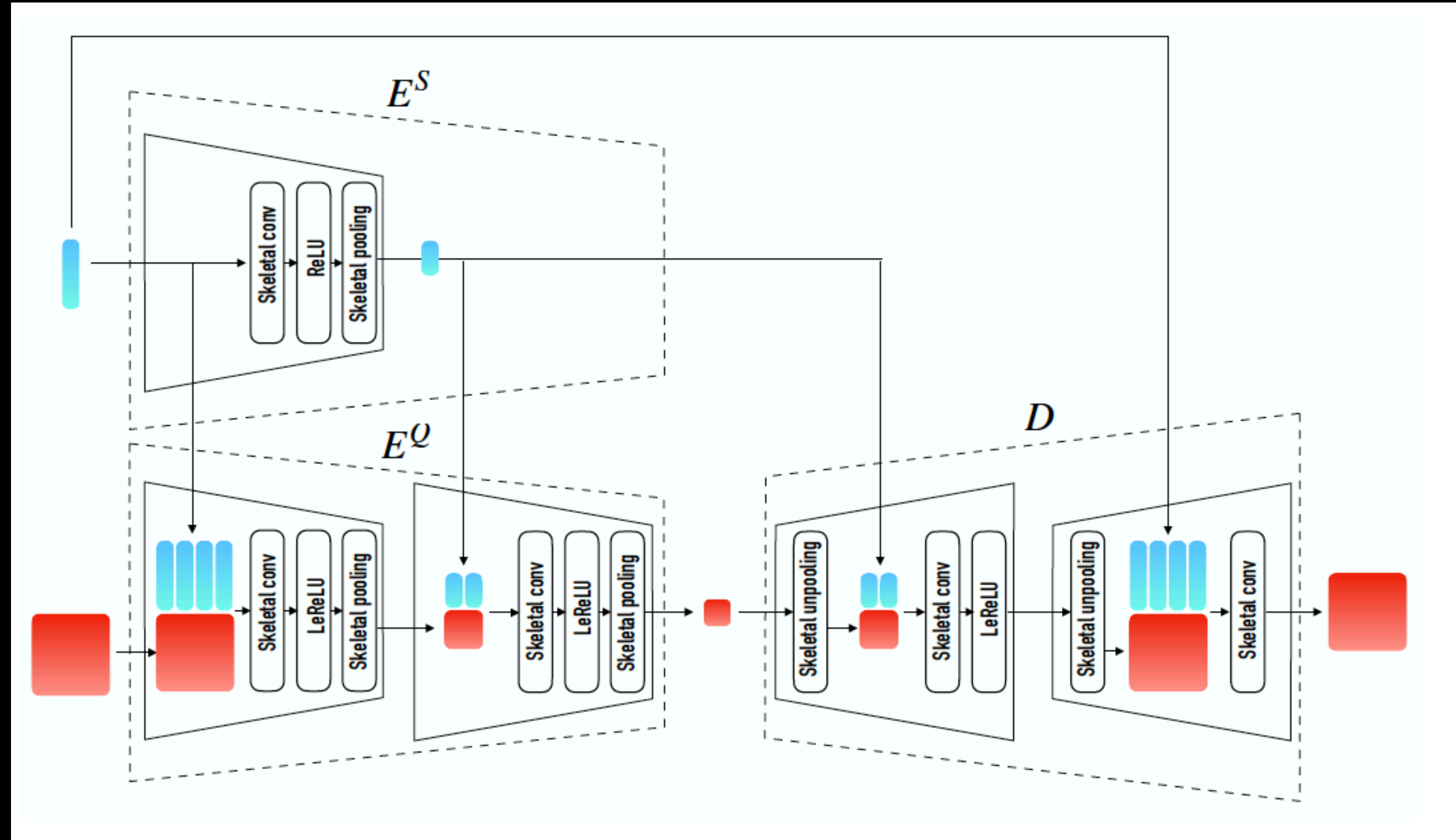
Deep learning overview



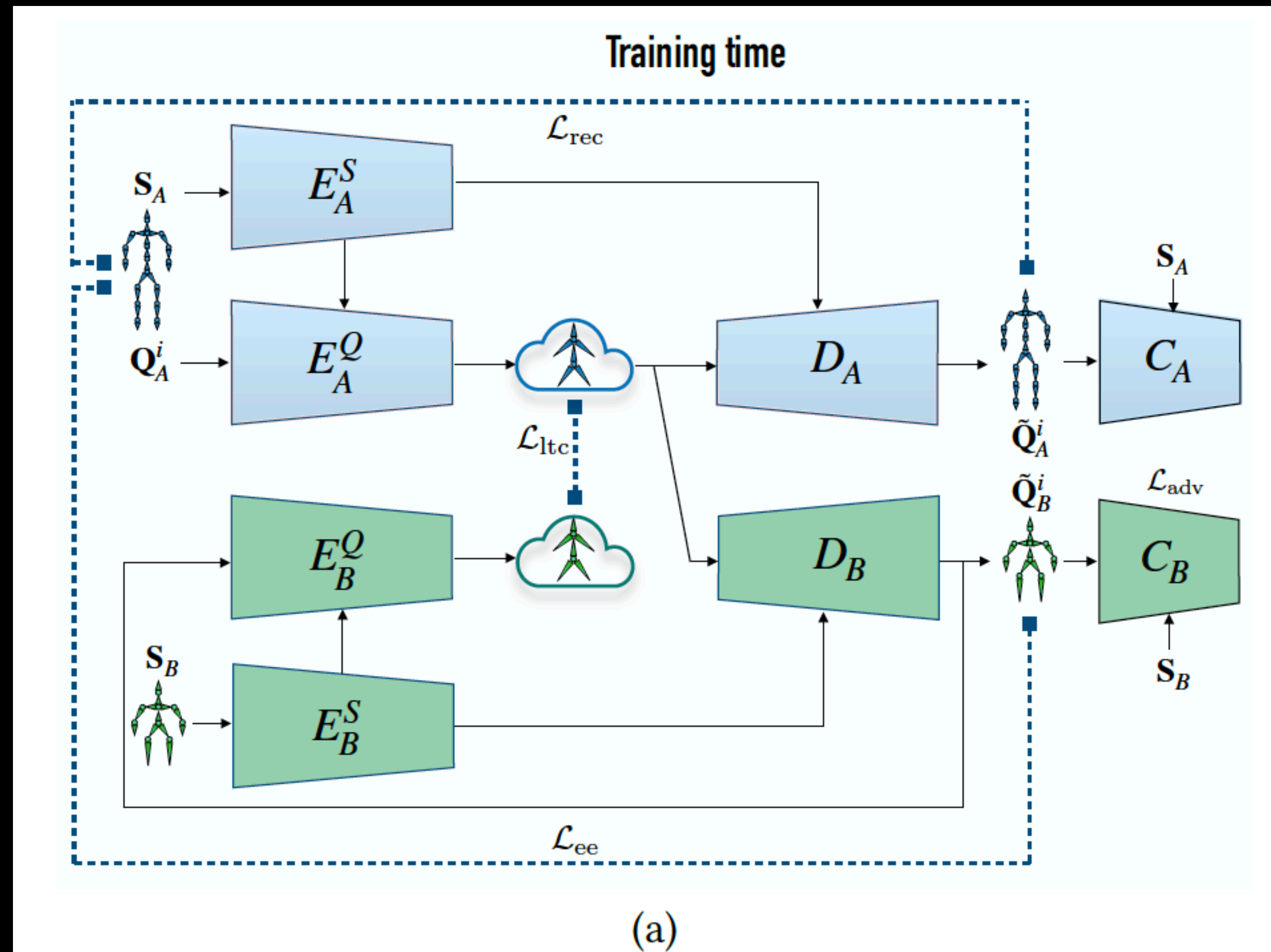
Deep learning overview



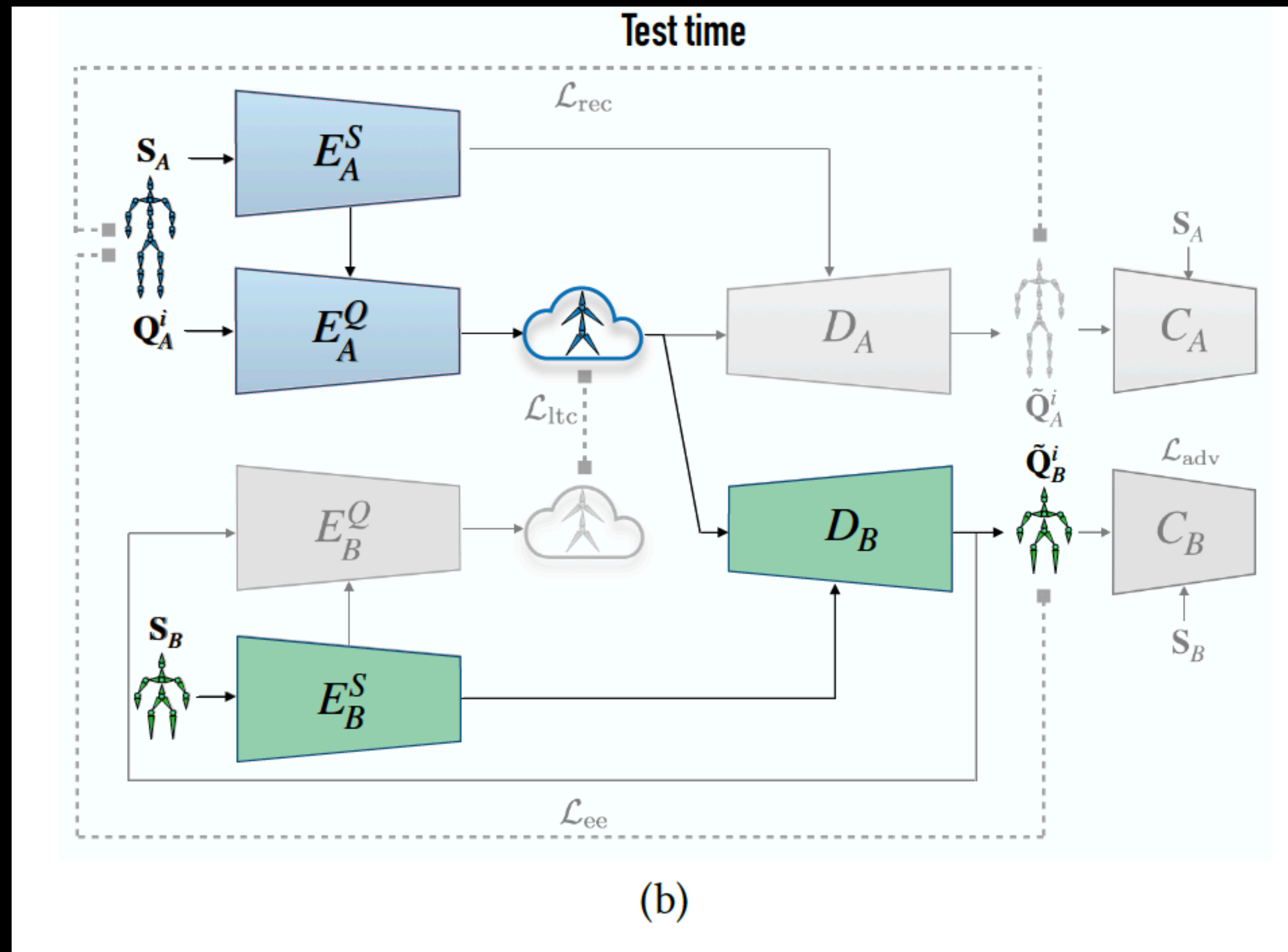
Overall Approach



Training



Creating Motion



Summary

- Motion retargeting
 - Still complex job
 - With the same structure - relatively easy
 - With different structure
 - Rigging and mapping is still complex
 - Many methods are proposed and gets better